



Cora Chen (Attempt 1 of

1)

Test Information

QUESTION 1: MULTIPLE CHOICE

10

out of 10 points

Consider a refrigeration cycle for the following specific case

1. Evaporation temperature 5°F
2. Condensing temperature 86°F
3. Liquid comes to the expansion valve without subcooling
4. Vapor entering the compressor is saturated



Select the optimal refrigerant from the list below: Assume all refrigerants follow the Peng-Robinson Equation of State

Given Ethyl Chloride

Answer:

Correct Ethyl Chloride

Answer:

QUESTION 2: CALCULATED NUMERIC

10

out of 10 points

The **Boyle** temperature of a gas is defined as the temperature at

which the derivative $\left. \frac{\partial (PV)}{\partial P} \right|_T = 0$ as $P \rightarrow 0$. Calculate the

Boyle temperature of Carbon Dioxide. For this calculation assume that Carbon Dioxide follows the Van der Waals Equation



of State. Express your result in $^{\circ}\text{C}$ and round your numerical answer to **a whole number**.

Given Answer: 753

Correct Answer: 754

Answer range +/- 0 (754 - 754)

QUESTION 3: CALCULATED NUMERIC

10

out of 10 points

Calculate the fugacity of CO_2 at temperature of $T=50^\circ\text{C}$ and pressure $P=100\text{ atm}$ using the compressibility factor data below. Express your result in **atm** and round your numerical answer to **a single decimal digit**.



Pressure (atm)	Compressibility Factor
5	0.990
10	0.970
20	0.930
30	0.885
40	0.830
50	0.770
60	0.710
70	0.650
80	0.600
90	0.525
100	0.430

Given Answer:  65.4

Correct Answer:  65.3

Answer range +/- 0.3 (65.0 - 65.6)

QUESTION 4: CALCULATED NUMERIC

0

out of 10 points

1 lbmole of CO_2 is compressed isothermally from $P_1=1\text{ atm}$ to $P_2=500\text{ psia}$ at a temperature of 70°F . Assume that CO_2 follows the Redlich-Kwong-Soave Equation of State. Calculate the heat that must be removed to maintain the compression isothermal. Express your result in **kJ** and round your numerical answer to **a single decimal digit**.



Given Answer:  3406.4

Correct Answer:  -4,458.7

Answer range +/- 1 (-4459.7 - -4457.7)

QUESTION 5: CALCULATED NUMERIC

10

out of 10 points

Consider the equilibrium reaction $\text{N}_2 + 3\text{H}_2 = 2\text{NH}_3$. Suppose we start with 10 moles of N_2 and carry out the reaction at temperature 400°C and pressure 150 bar . Under these conditions the reaction mixture is described by the Peng-Robinson Equation of State. What amount of hydrogen maximizes the mole fraction of ammonia in the reaction mixture. Express your result in **moles** and round your numerical answer to **a whole number**.



Given Answer: 30

Correct Answer: 30

Answer range +/- 1 (29 - 31)

QUESTION 6: CALCULATED NUMERIC

0

out of 10 points

The **Wohl's** Equation of State is given by the formula:

$$P + \frac{a}{V(V-b)} = \frac{RT}{V-b} + \frac{c}{V^3}$$

Using this Equation of State calculate the Molar Enthalpy of Ammonia at temperature: $T = 25^\circ\text{C}$ and $P = 6\text{ bar}$. Under these conditions Ammonia is in the vapor phase



Express your result in **kJ/mol** and round your numerical answer to **a single decimal digit**.

Given Answer: 2.9

Correct Answer: -45.9

Answer range +/- 0.2 (-46.1 - -45.7)

QUESTION 7: CALCULATED NUMERIC

0

out of 10 points

Calculate the **HIGH** inversion temperature of the Joule-Thomson coefficient for Nitrogen at 100 *atm*, assuming the Beattie-Bridgeman Virial Equation of State. The inversion temperature is defined as the the temperature at which the Joule-Thomson coefficient is equal to zero. Express your result in **K** and round your numerical answer to **a whole number**.

The constants in the virial equation of state have the values shown in the table below, when pressure is in *atm*, the density in



g/cm³ and the Universal Gas Constant in *cm³ · atm / (g · K)*.

Ao	1528.600000
a	0.274800
Bo	1.643000
b	-0.923500
c	2.200000E+06
d	1.413166

Given Answer: 44

Correct Answer: 573

Answer range +/- 0 (573 - 573)

QUESTION 8: CALCULATED NUMERIC

10

out of 10 points

Dry air at 75 *atm* and 300 *K* is to be cooled down to 100 *K* in a heat exchanger using nitrogen that enters the heat exchanger at 1 *atm* and 80 *K*. The flow rate of air is 1 *mol/s* and that of nitrogen 2.15 *mol/s*. Calculate the temperature of the nitrogen as it leaves the heat exchanger. Express your result in **K** and round your numerical answer to **a single decimal digit**.



Note: Assume Air and Nitrogen follow the Peng-Robinson Equation of State

Given Answer: 247.3

Correct Answer: 243.5

Answer range +/- 0.1 (243.4 - 243.6)

QUESTION 9: CALCULATED NUMERIC

0

out of 10 points

Calculate the solubility of Nitrogen in water at pressure $P=800$ *atm* and temperature $T=-50$ $^{\circ}\text{C}$. Assume that the vapor phase of Nitrogen under these conditions follows the Beattie-Bridgeman virial equation. The constants in the virial equation have the values shown in the table below, when pressure is in *atm*, the density in g/cm^3 and the Universal Gas Constant in $\text{cm}^3 \cdot \text{atm} / (\text{g} \cdot \text{K})$. Express your result in **mg**(of Nitrogen)/**L**(of pure water), and round your numerical answer to **a single decimal digit**.



Ao	1528.600000
a	0.274800
Bo	1.643000
b	-0.923500
c	2.200000E+06
d	1.413166

Given Answer: 12.3

Correct Answer: 342,657

Answer range +/- 100 (342557 - 342757)

QUESTION 10: CALCULATED NUMERIC

0

out of 10 points

A closed tank with volume $V_t = 10 \text{ ft}^3$ is filled with dry saturated steam at pressure $P_o = 265 \text{ psia}$. By removing a certain amount of heat (via cooling) 25% of steam is condensed. Calculate how much heat is removed to cause the condensation.



Express your result in **kJ** and round your numerical answer to **a whole number**.

Given Answer: 1409

Correct Answer: -1,168

Answer range +/- 2 (-1170 - -1166)

FEEDBACK AND NOTES FOR ATTEMPT